

Chapter 8: Polar Coordinates and Parametric Equations

Name: _____

Date: _____

Instructions: Answer all questions. Show complete work and justify your steps. Give *exact* values unless otherwise specified. Use radians unless degrees are indicated. Each problem begins on a new page.

1. Convert each coordinate pair as indicated. Give a principal polar angle with $0 \leq \theta < 2\pi$. (10)

(a) Rectangular to polar: $(-4, -4\sqrt{3})$.

(b) Polar to rectangular: $(-8, \frac{9\pi}{4})$.

2. Graph and analyze the line defined by $r = \frac{4}{2 \cos \theta - 5 \sin \theta}$. Convert to rectangular form and report slope, x - and y -intercepts, and provide a clear sketch. (12)

3. Graph each curve and answer the indicated questions.

(12)

(a) $r = \sin(2\theta)$

(b) $r = \sin(3\theta)$

(c) $r = \sin(4\theta)$

(d) $r = \sin(5\theta)$

- (e) Generalize: describe the graph of $r = \sin(k\theta)$ for any integer k (number of petals and petal length; indicate when petals repeat).

4. Let $z_1 = 3\left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$ and $z_2 = 5\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$. Compute z_1z_2 and $\frac{z_1}{z_2}$ in both polar and rectangular form. (12)

5. Compute all of the fifth roots of the complex number $a = -1 - i$. Express your answers in polar form and plot their arguments on the unit circle, labeling one exact rectangular form. (12)

6. Consider the parametric curve $x = 4 \cos t - 1$, $y = 3 \sin t$ for $0 \leq t \leq 2\pi$. (12)
- (a) Sketch the curve and use arrows to indicate the direction of motion as t increases.

- (b) Eliminate t to obtain a rectangular equation and state the points corresponding to $t = 0$ and $t = \pi$.

7. Consider the parametric equations (8)

$$x = \frac{4}{1+t^2}, \quad y = \frac{4t}{1+t^2}, \quad -\infty < t < \infty.$$

- (a) Eliminate the parameter to obtain a rectangular equation and identify the curve and its key features (center, radius).
- (b) State the points corresponding to $t = 0$ and $t \rightarrow \infty$, and indicate the orientation of traversal as t increases from 0 to $+\infty$.

8. An object moves in uniform circular motion modeled by (14)

$$x = 4 \cos(3t), \quad y = 4 \sin(3t), \quad t \text{ in seconds.}$$

- (a) Describe the path (center, radius), the position at $t = 0$, the orientation of motion, and the time for one full revolution.
- (b) Suppose the speed doubles while the center and radius remain the same. Give new parametric equations consistent with the same starting point and orientation.
- (c) Eliminate the parameter to obtain a rectangular equation for the path and state the direction of traversal for increasing t .